MIL-A-28745A(EC)
4 September 1970
SUPERSEDING
MIL-A-28745(EC)
21 April 1970
(See 6.5)

MILITARY SPECIFICATION

ANTENNA ASSEMBLY, OMNIDIRECTIONAL, LIGHT WEIGHT

1. SCOPE

1.1 This specification covers the parameters, design, fabrication, and tests of a light weight omnidirectional shipboard IFF antenna suitable for use with IFF transponders, IFF test sets and ISLS application. The object of this specification is to provide a light weight omnidirectional antenna for shipboard use that is compact, reliable, simple and inexpensive. The antenna shall be electrically and mechanically equivalent to, and interchangeable with, the existing AS-177 type omnidirectional IFF antennas.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17/80 - Cable, Radio Frequency, Coaxial, RG-219/U.
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-Q-9858 - Quality Program Requirements.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

SHIPS

REC49152 - Connector, Plug UG-154B/U.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

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(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 General requirements. The equipment shall be in accordance with the requirements of MIL-E-16400, except as hereinafter specified. Screws and screw-thread devices shall be in accordance with Handbook H28.

3.1.1 Painting. The enamel (air-drying speed) shall be Type III (baked), in accordance with MIL-E-15090, except where the application of this enamel would degrade the antenna performance.

3.1.2 Marking. The antenna radome shall be prominently marked "DO NOT PAINT" in a permanent manner.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.3 Reliability. Reliability requirements, testing or predictions are not applicable.

3.4 Design requirements.

3.4.1 Electrical.

3.4.1.1 Frequency coverage. The antenna shall be capable of transmitting at 1030 ±2 megahertz (MHz) and reception at 1090 ±2 MHz. The electrical characteristics shall be optimum at 1030 and 1090 MHz.

3.4.1.2 Power capacity. The antenna shall be capable of transmitting RF energy of at least 5.0 kilowatt (kW) peak power at a one percent duty cycle without breakdown under service conditions at the transmission frequency of 1030 ±2 MHz.

3.4.1.3 Polarization. The antenna shall transmit and receive vertical polarization RF energy on any azimuth angle.

3.4.1.4 Horizontal pattern. The horizontal or azimuth pattern shall be uniform with a maximum total variation in azimuth gain not greater than 1.5 dB.

3.4.1.5 Vertical pattern. The vertical beamwidth, as measured between the half power points, shall be 45 ±5 degrees. The maximum gain in the vertical shall lie between 2 and 4 degrees elevation in the vertical plane.

3.4.1.6 Impedance. The impedance shall be 52 ohms nominally over the 1030 to 1090 MHz band. Mismatch of the antenna and its associated filter shall be not greater than 3 dB over this frequency band.

3.4.1.7 Gain. The average maximum gain shall be not less than 2.8 dB over that of isotropic source over the 1030 to 1090 MHz band.

3.4.1.8 Losses. The housing or radome shall cause not more than 0.5 dB power loss over the frequency band, as measured by tests acceptable to the command or agency concerned.

3.4.1.9 Filter. A rejection filter shall be provided as an integral part of the antenna with the following characteristics:

(a) Rejection ratio of at least 40 dB from 2450 to 3530 MHz.
(b) Rejection ratio of at least 27 dB from 3530 to 3759 MHz.
(c) Mismatch not greater than 1.5 dB over the 1030 to 1090 MHz band.

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3.4.1.10 Blown fuse indicators. Not applicable.

3.4.1.11 Synchros. Not applicable.

3.4.1.12 Motors and generators. Not applicable.

3.4.2 Mechanical.

3.4.2.1 Electrical connection. The antenna shall be provided with a receptacle to which an LC series plug UG-154B/U in accordance with REC49152 shall be connected. The UG-154B/U plug is for RG-219/U cable, in accordance with MIL-C-17/80, and shall be provided as part of the antenna.

3.4.2.2 Mounting. Provision shall be made for mounting of the antenna in a vertical position, on a $1\frac{1}{2}$ inch extra heavy pipe, threaded for at least $2\frac{1}{2}$ inches with $1\frac{1}{2}$-$11\frac{1}{2}$ National Pipe Straight Mechanical (NPSM) thread in accordance with Handbook H28. Two clamp screws shall also be provided to prevent the antenna from moving after it has been threaded on the pipe. The method of mounting the antenna shall be submitted to the command or agency concerned for approval (see 6.3).

3.4.2.3 Time meters. Time meters are not required.

3.4.2.4 Instruction plates. Instruction plates are not required (see 6.3).

3.4.2.5 Weight. Total weight of the antenna shall not exceed 8.0 pounds, including any ancillary mounts or mounting devices or adapters as may be supplied.

3.4.3 Environmental conditions. The equipment shall be capable of operation under the following conditions as specified in MIL-E-16400, or as modified herein:

(a) Temperature; Class 2, except the upper limit shall be +125°C.
(b) Humidity.
(c) Wind speed of up to 100 knots relative.
(d) Icing.
(e) Nuclear blast (calculations only).

3.4.3.1 Shock. The antenna shall be capable of withstanding the shock test for Grade A, Type A, Class I, lightweight equipment as specified in MIL-S-901.

3.4.3.2 Degree of enclosure. The antenna shall be watertight as defined by MIL-STD-108.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The supplier shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under this contract. The quality control system shall be in accordance with MIL-Q-9858. A quality program plan, as it would apply to the requirements of this contract, shall be prepared prior to first article inspection (see 6.3). If first article inspection is not required, the quality program plan shall be prepared prior to production inspection. This program plan shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection

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positions shall be clearly identified on the flow charts and cross-referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the quality control system or the quality program plan shall be submitted to the cognizant Government Inspector for approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The supplier shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Each test shall be identified in accordance with 4.2 as to classification (see 6.3).

4.1.3.1 Procedures for inspections, tests, and demonstrations shall contain step-by-step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
    (1) Production inspection (see 4.4.1).
    (2) Production control inspection (see 4.4.2).
    (3) Environmental tests (see 4.4.3).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection.

Examination or test	Test method paragraph	Number of sample units to be inspected	Number of failures
Surface examination	4.5.1	one	0
Operating test	4.5.2	one	0
Weights and dimensions	4.5.3	one	0
Enclosure test 1/	4.5.4	one	0
Temperature and humidity tests	4.5.5	one	0

1/ Enclosure test shall be performed subsequent to shock and vibration tests.

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TABLE I. First article inspection. (Continued)

Examination or test	Test method paragraph	Number of sample units to be inspected	Number of failures
Shock and vibration tests	4.5.6	one	0
Salt spray test	4.5.7	one	0
Wind velocity test	4.5.8	one	0
Icing examination	4.5.8	one	0
Nuclear blast test	Calculations only		

4.3.1 Verification of characteristics. In addition, after the first article tests have been performed, the following characteristics of the antenna (previously determined) shall be checked to determine what changes may have taken place:

(a) Horizontal pattern.
(b) Vertical pattern.
(c) Gain.
(d) Radome losses.
(e) Filter rejection.

4.3.2 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data. The first article inspection report shall indicate specification compliance limits as well as actual performance test results. The supplier shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the requirements of table II.

TABLE II. Production inspection.

Inspection	Reference paragraph
Surface examination	4.5.1
Operating test	4.5.2

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

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TABLE III. Production control inspection.

Inspection	Reference paragraph	AQL percent
Weights and dimensions	4.5.3	6.5
Enclosure test	4.5.4	6.5
Operating test	4.5.2	6.5

4.4.2.1 In addition, the verification of the characteristics of the antenna as specified in 4.3.1 shall be made after production control testing.

4.4.3 Environmental tests. Environmental tests shall be as listed in table IV and shall be performed on units which have been subjected to and passed production inspection. Environmental test procedures shall be approved in accordance with 4.1.3.

TABLE IV. Environmental tests.

Inspection	Reference paragraph
Temperature and humidity tests	4.5.5
Shock and vibration tests	4.5.6
Salt spray test	4.5.7

4.4.3.1 In addition, the verification of the characteristics of the antenna as specified in 4.3.1 shall be made after environmental testing.

4.4.3.2 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

(a) One sample from the first 25 production units fabricated
(b) One sample from each successive 75 units produced, or fraction thereof

4.4.3.3 Nonconforming production control or environmental sample units. If a sample unit fails the test specified in 4.4.2 or 4.4.3, the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed production control and environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Inspection procedures.

4.5.1 Surface examination. Each antenna assembly shall be examined for completeness, workmanship, assembly and fit, mechanical mounting, electrical connections, materials, parts, and finish to determine that material and workmanship requirements have been met.

4.5.2 Operating test. Each antenna assembly shall be assembled by the supplier and shall be given all performance tests necessary to confirm that the antenna meets the performance requirements specified herein. As a minimum the operating test shall include power capacity tests, impedance and gain tests, and filter rejection tests to demonstrate compliance with 3.4.1.2, 3.4.1.6, 3.4.1.7, and 3.4.1.9.

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4.5.3 Weights and dimensions. Weights and dimensions shall be determined in accordance with MIL-E-16400 to verify compliance with 3.4.2.5.

4.5.4 Enclosure test. Watertight testing shall be performed in accordance with MIL-STD-108 (water repellent test) to assure compliance with 3.4.3.2.

4.5.5 Temperature and humidity. The antenna assembly shall be subjected to nonoperating and operating condition temperature tests (Class 2, but with upper temperature limit of 125°C) and humidity tests in accordance with MIL-E-16400 to determine compliance with 3.4.3.

4.5.6 Shock and vibration. The antenna assembly shall meet the shock requirements of Grade A, Class I, Type A, lightweight equipment of MIL-S-901 and the vibration requirements of Type 1 of MIL-STD-167, as specified in MIL-E-16400. The antenna shall be in the operating condition during and after the shock and vibration tests. The method of mounting the antenna assembly for testing shall be subject to the approval of the command or agency concerned.

4.5.7 Salt spray test. The salt spray test shall be made in accordance with MIL-E-16400.

4.5.8 Wind velocity and icing. In addition to the weight of the antenna, the antenna shall be designed to withstand the following simulated load conditions:

(a) Wind speed of up to 100 knots relative.
(b) Ice loads of 4.5 pounds per square foot.

The methods of simulation of these load conditions shall be subject to the approval of the command or agency concerned.

4.6 Inspection of preparation for delivery. Examinations shall be conducted to insure conformance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. The equipment, accessories, technical publications and repair parts shall be preserved and packaged level A or C, packed level A, B, C, and marked as specified in the contract or order in accordance with MIL-E-17555 (See 6.2).

6. NOTES

6.1 Intended use. This equipment is a lightweight, omnidirectional shipboard IFF antenna assembly suitable for use with IFF transponders, IFF test sets, and ISLS application.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1).

6.3 Contact data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

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(a) Technical manuals including operating instructions are to be provided with each equipment if required in the contract or order.
(b) Method of mounting the antenna (see 3.4.2.2).
(c) Contractor's quality program plan (see 4.1.1).
(d) Test procedures (see 4.1.3).
(e) First article inspection report (see 4.3.2).
(f) Failure reports (see 4.4.3.3).

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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